Task 1: Data Quality and Initial Exploratory Assessment
Marketing Campaign Performance Analysis

July 29, 2026

Tools

- Python with VSCode

Task 1

Data Quality and Initial Exploratory Assessment

1.1 Data Quality Assessment

Prior to conducting business intelligence analysis, the advertising campaign dataset was evaluated to ensure its quality, reliability, and suitability for analysis. The objective of this assessment is to identify potential data quality issues, validate the performance metrics used throughout the project, examine unusual observations, and establish a set of baseline metrics that will guide subsequent business analysis.

The dataset contains daily advertising campaign performance across multiple platforms, regions, target audiences, product categories, and creative types during the first quarter of 2024.

1.1.1 Data Type

The data types of all variables were first examined to verify that each column was stored using an appropriate data type. As shown in *Fig. 1.1*, numerical variables were stored as either `int64` or `float64`, categorical variables as `object`, the `Date` column as `datetime64[ns]`, and the `Is_Competitive_Event` variable as a boolean.

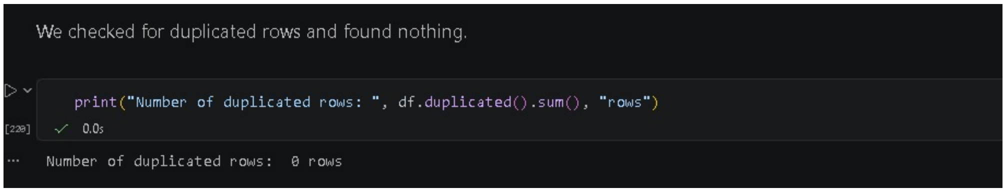
```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3240 entries, 0 to 3239
Data columns (total 17 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Date                   3240 non-null   datetime64[ns]
 1   Platform               3240 non-null   object
 2   Campaign               3240 non-null   object
 3   Region                 3240 non-null   object
 4   Spend                  3240 non-null   float64
 5   CPM                    3240 non-null   float64
 6   Impressions            3240 non-null   int64
 7   Frequency              3240 non-null   float64
 8   Clicks                 3240 non-null   int64
 9   Purchases              3240 non-null   int64
 10  Revenue                3220 non-null   float64
 11  Product_Category       3240 non-null   object
 12  Target_Audience        3240 non-null   object
 13  Creative_Type          3240 non-null   object
 14  Video_Completion_Rate  1228 non-null   float64
 15  Customer_LTV           3240 non-null   float64
 16  Is_Competitive_Event   3240 non-null   bool
```

Figure 1.1: Checking datatypes of the dataframe

No inconsistencies in data types were identified; therefore, no type conversions were required.

1.1.2 Duplicate Records

The dataset was examined for duplicate observations.



We checked for duplicated rows and found nothing.

```
print("Number of duplicated rows: ", df.duplicated().sum(), "rows")
```

[220] 0.0s

```
Number of duplicated rows:  0 rows
```

Figure 1.2: Checking for duplicate records

No duplicate records were identified as shown in *Fig. 1.2*.

1.1.3 Data Completeness

The dataset was examined for missing values to determine whether incomplete observations could affect subsequent analyses.

- **Revenue**

  A sample of these datapoints are shown in *Fig. 1.3*. It can be seen that twenty observations (approximately 0.6% of the dataset) contained missing revenue values.

2. Revenue

There seems to be some missing values for the Revenue column as well. This column is very important as it is used as the basis for the KPI metrics like ROAS, CPA, etc..

```
df[df['Revenue'].isna()].head()
```

[218] 0.0s Python

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate	Customer_LTV	Is_Competiti
194	2024-01-06	Google	Google_Athletes_Search_Protein_001	Northeast	1225.52	18.09	67745	2.27	1236	35	NaN	Protein	Athletes	Search	NaN	940.8	
589	2024-01-17	Google	Google_Athletes_Search_Protein_001	South	978.48	15.24	64204	2.03	979	32	NaN	Protein	Athletes	Search	NaN	940.8	
1413	2024-02-09	FB	FB_FitnessEnth_Carousel_WeightLoss_001	South	1909.27	12.47	153133	3.61	823	24	NaN	WeightLoss	FitnessEnth	Carousel	NaN	448.5	
1658	2024-02-16	FB	FB_Athletes_Video_Protein_001	Northeast	974.65	12.55	77649	2.08	505	16	NaN	Protein	Athletes	Video	0.4413	772.8	
1661	2024-02-16	FB	FB_Athletes_Image_Preworkout_002	South	913.94	12.55	72812	4.25	243	6	NaN	Preworkout	Athletes	Image	NaN	538.2	
1682	2024-02-16	TT	TT_Athletes_Video_Protein_001	Northeast	920.33	10.46	87985	2.92	513	20	NaN	Protein	Athletes	Video	0.6728	604.8	
1738	2024-02-18	FB	FB_FitnessEnth_Carousel_WeightLoss_001	Northeast	2696.87	15.09	178704	2.17	881	28	NaN	WeightLoss	FitnessEnth	Carousel	NaN	448.5	

Figure 1.3: Sample datapoints with missing revenues.

  Because these represented a very small proportion of the data and no systematic pattern of missingness was identified, the corresponding observations were removed as shown in *Fig. 1.4* prior to analysis.

- **Video Completion Rate**

  A large number of missing values were initially observed in the *Video Completion Rate* variable. Around 62% of the total dataset!

  Further investigation revealed that this metric is only applicable to **campaigns using video advertisements**. After restricting the dataframe to video creatives, **only 212 missing observations remained** as shown in *Fig. 1.6*.

Because the proportion of missing values is very small, removing these observations is unlikely to introduce significant bias or substantially reduce the available information. Consequently, these rows will be removed from the dataset for analyses involving the `Revenue` variable.

```python
print("BEFORE DROPPING: Number of rows with missing revenue:",len(df[df['Revenue'].isna()]))

df.dropna(subset=["Revenue"], inplace=True)

print("AFTER DROPPING: Number of rows with missing revenue:",len(df[df['Revenue'].isna()]))
```

```
BEFORE DROPPING: Number of rows with missing revenue: 20
AFTER DROPPING: Number of rows with missing revenue: 0
```

Figure 1.4: Dropped datapoints with missing revenues.

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3240 entries, 0 to 3239
Data columns (total 17 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Date                   3240 non-null   datetime64[ns]
 1   Platform               3240 non-null   object
 2   Campaign               3240 non-null   object
 3   Region                 3240 non-null   object
 4   Spend                  3240 non-null   float64
 5   CPM                    3240 non-null   float64
 6   Impressions            3240 non-null   int64
 7   Frequency              3240 non-null   float64
 8   Clicks                 3240 non-null   int64
 9   Purchases              3240 non-null   int64
 10  Revenue                3220 non-null   float64
 11  Product_Category       3240 non-null   object
 12  Target_Audience        3240 non-null   object
 13  Creative_Type          3240 non-null   object
 14  Video_Completion_Rate  1228 non-null   float64
 15  Customer_LTV           3240 non-null   float64
 16  Is_Competitive_Event   3240 non-null   bool
```

Figure 1.5: `video_completion_rate` missing values

Consequently, these missing values were treated as **structurally missing** rather than data quality issues, and analyses involving this variable will use only the relevant subset of campaigns.

1.1.4 Logical Consistency Checks

Several validation checks were performed to identify logically inconsistent observations.

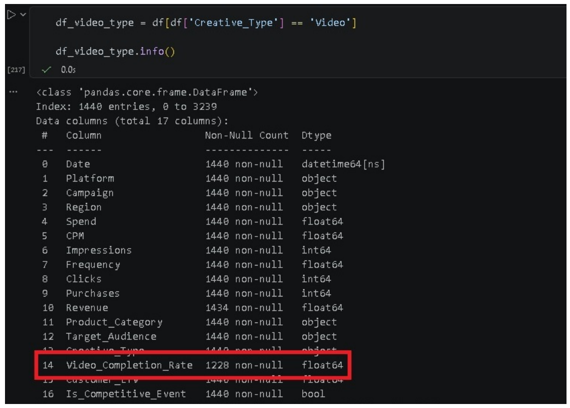
0 Clicks with non-zero purchase

I noticed that there were rows with 0 clicks but has non-zero Purchases, which I thought was weird at first. How are they able to purchase without clicking the ad.

Apparently, this happens in real world scenarios and they are called **view-through conversions**. People see the ad, they don't click it then later that evening, they remembered the brand and typed the website into google and bought the product.

But, at this point, I was wondering how they know which ad actually influenced the purchased without click or what if the buyer interacted with multiple ads before purchasing(cross-channel conversions). Apparently, this is related to **attribution**, and there are multiple attribution models like **last_click attribution**, **first-click attribution**, **linear_attribution**, etc.. which is not part of this data cleaning process.

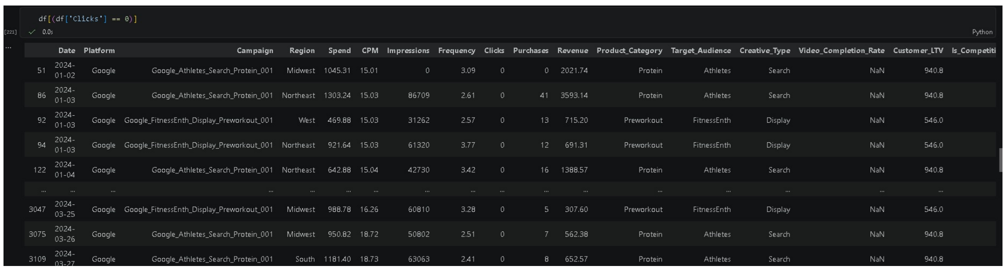
So, for now, I'll just assume that this dataset was collected using a certain attribution model and leave it at that. This justifies the 0 clicks with non-zero purchase.



```
df_video_type = df[df['Creative_Type'] == 'Video']

df_video_type.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 1440 entries, 0 to 3239
Data columns (total 17 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Date                   1440 non-null   datetime64[ns]
 1   Platform               1440 non-null   object
 2   Campaign               1440 non-null   object
 3   Region                 1440 non-null   object
 4   Spend                  1440 non-null   float64
 5   CPM                    1440 non-null   float64
 6   Impressions            1440 non-null   int64
 7   Frequency              1440 non-null   float64
 8   Clicks                 1440 non-null   int64
 9   Purchases              1440 non-null   int64
 10  Revenue                1434 non-null   float64
 11  Product_Category       1440 non-null   object
 12  Target_Audience        1440 non-null   object
 13  Creative_Type          1440 non-null   object
 14  Video_Completion_Rate  1228 non-null   float64
 15  Customer_LTV           1440 non-null   float64
 16  Is_Competitive_Event   1440 non-null   bool
```

Figure 1.6: DataFrame filtered to video creative types



```
df[(df['Clicks'] == 0)]
```

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate	Customer_LTV	Is_Competiti
51	2024-01-02	Google	Google_Athletes_Search_Protein_001	Midwest	1045.31	15.01	0	3.09	0	0	2021.74	Protein	Athletes	Search	NaN	940.8	
86	2024-01-03	Google	Google_Athletes_Search_Protein_001	Northeast	1303.24	15.03	86709	2.61	0	41	3593.14	Protein	Athletes	Search	NaN	940.8	
92	2024-01-03	Google	Google_FitnessEnth_Display_Preworkout_001	West	469.88	15.03	31262	2.57	0	13	715.20	Preworkout	FitnessEnth	Display	NaN	546.0	
94	2024-01-03	Google	Google_FitnessEnth_Display_Preworkout_001	Northeast	921.64	15.03	61320	3.77	0	12	691.31	Preworkout	FitnessEnth	Display	NaN	546.0	
122	2024-01-04	Google	Google_Athletes_Search_Protein_001	Northeast	642.88	15.04	42730	3.42	0	16	1388.57	Protein	Athletes	Search	NaN	940.8	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
3047	2024-03-25	Google	Google_FitnessEnth_Display_Preworkout_001	Midwest	988.78	16.26	60810	3.28	0	5	307.60	Preworkout	FitnessEnth	Display	NaN	546.0	
3075	2024-03-26	Google	Google_Athletes_Search_Protein_001	Midwest	950.82	18.72	50802	2.51	0	7	562.38	Protein	Athletes	Search	NaN	940.8	
3109	2024-03-27	Google	Google_Athletes_Search_Protein_001	South	1181.40	18.73	63063	2.41	0	8	652.57	Protein	Athletes	Search	NaN	940.8	

Figure 1.7: Sample datapoints with 0 clicks but non-zero purchase

0 Impressions with non-zero Cost-Per-Mile(CPM)

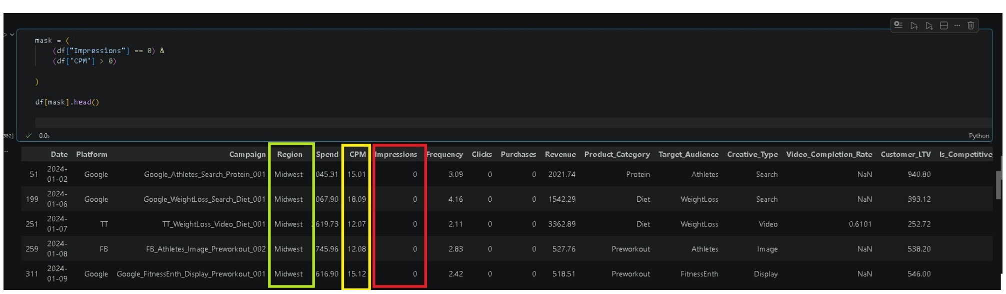
Cost Per Mille (CPM) measures the advertising cost required to generate 1,000 impressions and is commonly used to evaluate the cost efficiency of awareness campaigns.

$$\text{CPM} = \frac{\text{Spend}}{\text{Impressions}} \times 1000 \tag{1.1}$$

Because impressions appear in the denominator, CPM is undefined when the number of impressions is zero.

During the data quality assessment, 47 observations (approximately 1.45% of the dataset) were identified with `Impressions = 0` but non-zero with CPM values.

Furthermore, all of these observations originated from the Midwest region as shown in *Fig. 1.8*, suggesting a systematic reporting inconsistency rather than isolated data entry errors.



```
mask = (
    (df["Impressions"] == 0) &
    (df['CPM'] > 0)

)

df[mask].head()
```

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate	Customer_LTV	Is_Competitive
51	2024-01-02	Google	Google_Athletes_Search_Protein_001	Midwest	045.31	15.01	0	3.09	0	0	2021.74	Protein	Athletes	Search	NaN	940.80	
199	2024-01-06	Google	Google_WeightLoss_Search_Diet_001	Midwest	067.90	18.09	0	4.16	0	0	1542.29	Diet	WeightLoss	Search	NaN	393.12	
251	2024-01-07	TT	TT_WeightLoss_Video_Diet_001	Midwest	619.73	12.07	0	2.11	0	0	3362.89	Diet	WeightLoss	Video	0.6101	252.72	
259	2024-01-08	FB	FB_Athletes_Image_Preworkout_002	Midwest	745.96	12.08	0	2.83	0	0	527.76	Preworkout	Athletes	Image	NaN	538.20	
311	2024-01-09	Google	Google_FitnessEnth_Display_Preworkout_001	Midwest	616.90	15.12	0	2.42	0	0	518.51	Preworkout	FitnessEnth	Display	NaN	546.00	

Figure 1.8: Sample rows that has 0 impressions with non-zero CPM

Since these observations violate the definition of CPM and the 0 impressions value could affect the calculation of other performance metrics, such as CTR and CVR, they were removed from the dataset prior to analysis.

```
mask = (df["Impressions"] == 0) & (df["CPM"] > 0)

print("Rows before removal:", mask.sum())

df.drop(df[mask].index, inplace=True)

mask = (df["Impressions"] == 0) & (df["CPM"] > 0)
print("Rows after removal:", mask.sum())
[356]  ✓ 0.0s

... Rows before removal: 47
    Rows after removal: 0
```

Figure 1.9: Dropped Rows that has 0 impressions with non-zero CPM

1.1.5 Outlier Assessment

Boxplots were used to identify potential outliers across the numerical variables as shown in *Fig. 1.10*. Severe outliers were identified for the Spend column and will be investigated in this section.

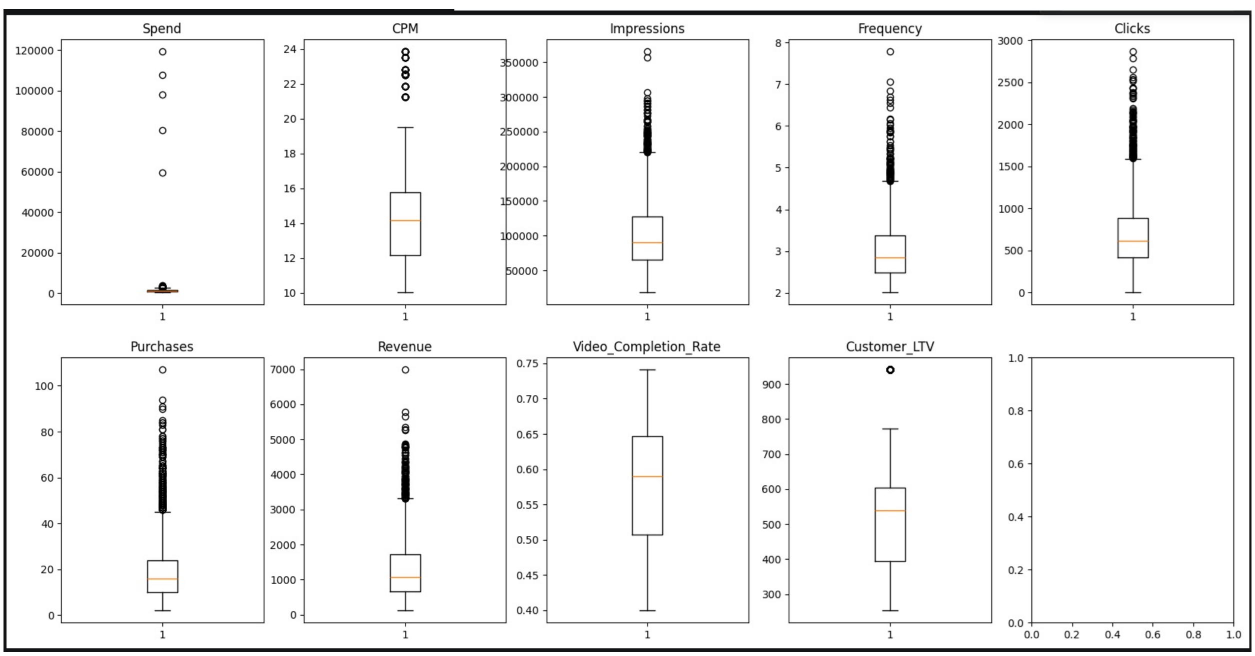


Figure 1.10: Boxplots for all numerical columns of the dataframe

`Spend` severe outliers investigation

First, I listed the 5 datapoints with severe `Spend` outliers as shown in *Fig. 1.12*. I noticed that in this 5 datapoints, the **CPM do not match the Ad Spend and Impression**. This tells me that there is a possibility that there are more rows with mismatched CPM.

```python
import numpy as np

df["Calculated_CPM"] = (df["Spend"] / df["Impressions"]) * 1000

# Compare with a tolerance of 0.01
matches = np.isclose(df["Calculated_CPM"], df["CPM"], atol=0.1)

mismatches = df[~matches]

print(f"Number of CPM mismatches: {len(mismatches)}")

mismatches[["Campaign","Spend", "Impressions", "CPM", "Calculated_CPM"]].sort_values(by = 'Spend', ascending=False)
```
[312] ✓ 0.0s

```
Number of CPM mismatches: 5
```

	Campaign	Spend	Impressions	CPM	Calculated_CPM
1286	TT_Athletes_Video_Protein_001	119315.0	115279	10.35	1035.010713
1425	Google_FitnessEnth_Display_Preworkout_001	107742.0	69131	15.58	1558.519333
2892	Google_Athletes_Search_Protein_001	97883.0	60421	16.20	1620.016220
2744	FB_FitnessEnth_Carousel_WeightLoss_001	80592.0	52013	15.49	1549.458789
923	Google_FitnessEnth_Display_Preworkout_001	59556.0	33683	17.68	1768.132292

Figure 1.11: Number of rows with CPM mismatch

So, I checked and found out that only these 5 rows have mismatched CPM as shown in *Fig. 1.11*, after dropping the rows from earlier(the ones with 0 impression,clicks and purchases).

```python
df.loc[outlier_masks["Spend"], "Spend"] /=100

df[outlier_masks["Spend"]].sort_values(by = 'Spend', ascending=False).head(5)
```
✓ 0.0s

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate
1286	2024-02-05	TT	TT_Athletes_Video_Protein_001	Northeast	1193.15	10.35	115279	2.97	705	21	1746.21	Protein	Athletes	Video	0.6891
1425	2024-02-09	Google	Google_FitnessEnth_Display_Preworkout_001	South	1077.42	15.58	69131	2.99	798	11	602.20	Preworkout	FitnessEnth	Display	NaN
2892	2024-03-21	Google	Google_Athletes_Search_Protein_001	West	978.83	16.20	60421	2.35	399	16	1325.27	Protein	Athletes	Search	NaN
2744	2024-03-17	FB	FB_FitnessEnth_Carousel_WeightLoss_001	West	805.92	15.49	52013	2.36	145	2	125.28	WeightLoss	FitnessEnth	Carousel	NaN
923	2024-01-26	Google	Google_FitnessEnth_Display_Preworkout_001	Midwest	595.56	17.68	33683	3.44	517	7	414.71	Preworkout	FitnessEnth	Display	NaN

Figure 1.12: Datapoints with severe `Spend` outliers

Then, I plotted the `Spend` boxplots of these Campaign Ads.
From this analysis, I noticed the following consistencies:

1. The reported CPM values(from datapoints with severe `Spend` outliers) were exactly one-hundredth of the values calculated from the corresponding Spend and Impressions.

2. The calculated CPM values do not match the 'Impressions' and 'Spend' columns, and this is limited only to the 5 reported rows with severe 'Spend' outliers.

3. Inspection of the affected campaigns showed that the unusually large Spend values occurred only on a single day and were not associated with competitive events.

Since the discrepancy was systematic (a factor of 100) and limited to these five observations, the Spend values were divided by 100 to restore consistency with the reported CPM values.

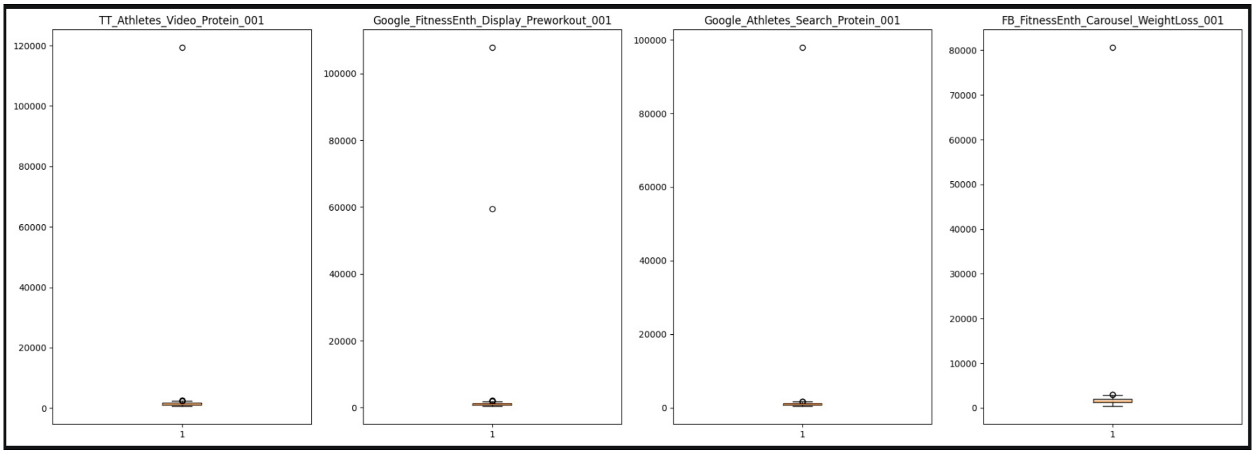


Figure 1.13: `Spend` boxplots for campaign ads with severe `Spend` outliers

1.1.6 Data Cleaning Summary

The data cleaning process consisted of:

- Verified variable data types
- Verifying duplicate observations
- Removing observations(20 datapoints) with missing revenue.
- Retaining structurally missing Video Completion Rate values.
- Retaining valid view-through conversions.
- Removing observations(47 datapoints) with 0 clicks,impressions and purchases with non-zero purchase.
- Addressed severe `Spend` outliers.

Overall, the dataset was determined to be sufficiently complete and reliable for subsequent analysis.

1.2 Business Context

The company is currently underperforming, with an overall Return on Advertising Spend (ROAS) of 0.86, which falls below the target value of 1.2. To improve profitability while achieving a 40% increase in revenue, the marketing team must optimize its advertising strategy and budget allocation.

1.2.1 Challenge

The primary business challenges include:

- Overall ROAS is substantially below the target value.

- Campaign performance varies considerably across advertising platforms, creative types, and product categories.
- Weekly ROAS has shown a declining trend, indicating decreasing advertising efficiency.
- Regional performance differences remain unclear.
- The impact of competitive events on campaign performance is uncertain.
- The effectiveness of different creative types and audience segments has not yet been established.
- Advertising spend must be reduced in underperforming areas while maintaining revenue growth.

1.2.2 Performance Metrics

We need numbers to evaluate campaign effectiveness. In that regard, four standard digital marketing performance metrics were considered.

Metric	Formula
CTR	Clicks / Impressions
CPC	Spend / Clicks
CVR	Purchases / Clicks
ROAS	Revenue / Spend

Table 1.1: Calculated performance metrics.

These metrics provide standardized measures of audience engagement, advertising efficiency, conversion effectiveness, and financial return.

1.3 Initial Exploratory Assessment

1.3.1 Business Overview

Following data cleaning, the overall Return on Advertising Spend (ROAS) is 0.96, which is below the company's target of 1.20. This indicates that the current advertising strategy is not generating sufficient revenue relative to advertising expenditure. Performance also varies across campaigns, platforms, creative types, product categories, and target audiences, suggesting opportunities to improve budget allocation and campaign effectiveness.

Since improving ROAS is the primary business objective, the subsequent analysis focuses on identifying the factors associated with both high- and low-performing advertising campaigns.

1.3.2 Key Performance Metrics

To evaluate advertising performance, the following baseline metrics were selected.

Metric	Business Relevance
Spend	Measures advertising investment and budget allocation.
Revenue	Measures the financial return generated by advertising activities.
ROAS	Primary indicator of advertising efficiency and campaign profitability.
CTR	Measures audience engagement with advertisements.
CPC	Measures the cost required to generate user engagement.
CVR	Measures the effectiveness of converting engaged users into customers.

Table 1.2: Baseline performance metrics used throughout the analysis.

Together, these metrics provide a comprehensive view of advertising performance, covering investment, engagement, conversion, and financial return.

1.3.3 Preliminary Insights

I created a power BI dashboard(file is inside github repository) to summarize the ROAS to gives us a main idea of the campaign ad ROAS across different categorical variables.

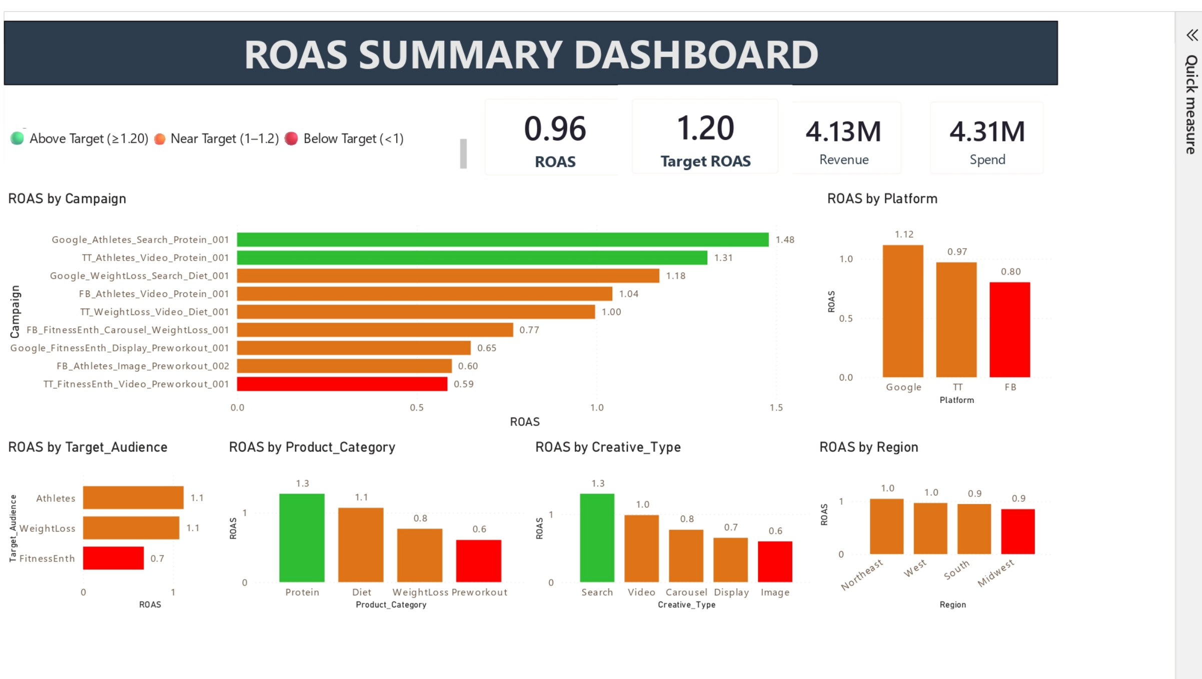


Figure 1.14: ROAS Dashboard Summary

The executive dashboard provides an initial overview of advertising performance prior to detailed analysis. Several observations can be made:

- Overall ROAS (0.96) remains below the business target of 1.20.

- Campaign performance varies substantially, with only a few campaigns exceeding the target ROAS.
- Google delivers the highest ROAS among advertising platforms, while Facebook records the lowest.
- Search creatives and Protein campaigns achieve the strongest advertising returns.
- Performance differences across regions are relatively small, whereas larger variation exists across creative types, product categories, and target audiences.

These observations establish a baseline understanding of current advertising performance and guide the subsequent analyses presented in Task 2.

Task 2

Performance Analysis

The exploratory analysis identified observable differences in performance metrics across multiple business dimensions, including advertising platform, region, creative type, product category, and target audience.

However, descriptive statistics alone cannot determine whether these observed differences represent genuine performance differences or are simply the result of random variation. Therefore, statistical significance testing was performed before drawing business conclusions.

2.1 Statistical Methodology

Preliminary assessment of the data showed that the distributions of several performance metrics, particularly ROAS, deviated significantly from normality. Such behavior is common in advertising datasets, where campaign performance is often positively skewed due to a small number of exceptionally high-performing campaigns.

Consequently, non-parametric statistical methods were adopted where appropriate.

The following statistical tests were used throughout the analysis:

1. **Kruskal–Wallis Test**

   A non-parametric alternative to one-way ANOVA used to determine whether the distribution of a performance metric differs significantly across multiple groups. This test was selected because the normality assumption required for ANOVA was not satisfied.

   Business Question Answered:

   > *Do the observed differences in campaign performance across business groups (e.g., platforms, regions, creative types, product categories, and target audiences) represent genuine differences, or are they likely due to random variation?*

2. **Levene's Test**

   Used to determine whether the variability (consistency) of a performance metric differs significantly across groups.

   Business Question Answered:

   > *Which business groups deliver more consistent advertising performance, and which exhibit greater variability or unpredictability?*

Together, these tests provide statistical evidence for evaluating differences in both performance and performance consistency across the business dimensions considered in this study.

2.2 Channel Performance Analysis

2.2.1 Statistical Results

Table 2.1 presents the descriptive statistics for the four key performance metrics across the three advertising platforms. Google achieved the highest average ROAS (1.106) and CTR (1.00%), while Facebook recorded the highest average CVR (2.87%). Google also achieved the lowest average CPC ($1.87), indicating lower acquisition costs than Facebook and TikTok.

Prior to hypothesis testing, the Shapiro–Wilk test indicated that the distributions of all performance metrics deviated significantly from normality ($p < 0.05$). Consequently, the Kruskal–Wallis test was used to compare platform performance.

For all four performance metrics (ROAS, CTR, CVR, and CPC), the Kruskal–Wallis test identified statistically significant differences among Facebook, Google, and TikTok ($p < 0.001$). Dunn's post-hoc analysis further confirmed that the majority of pairwise platform comparisons were statistically significant, indicating that advertising platform has a measurable influence on campaign performance.

Levene's test also identified significant differences in performance variability across platforms ($p < 0.05$), suggesting that some advertising platforms deliver more consistent campaign outcomes than others.

Table 2.1: Platform performance summary.

Platform	ROAS	CTR	CVR	CPC
Facebook	0.816	0.0057	**0.0287**	2.84
Google	**1.106**	**0.0100**	0.0254	**1.87**
TikTok	0.960	0.0063	0.0254	1.98